Kai Gu

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Education

Bocconi University	**Milan, Italy**
PhD in Marketing	2026 – Present
Bocconi University	**Milan, Italy**
MSc in Data Science	2024 – 2026
Zhejiang University	**Hangzhou, China**
BEcon in Economics	2020 – 2024

Research Interests

Substantive: Economics of AI, Economics of Digitization & Platforms, Media Economics
Methodological: Causal Inference, Computational Social Sciences, Multimodal Data

Working Papers

From Directory to Oracle: How AI Search Reshapes the Web's Attention Economy
Qiaoni Shi, Kai Zhu, and **Kai Gu**

Presented at: Bocconi Marketing Research Camp, May 2026; ISMS Marketing Science Conference, Lisbon, Portugal, June 2026

For two decades, the web's attention economy has rested on an implicit bargain: search engines help users find information and route traffic to the websites that produce it. AI search threatens to break this bargain by serving as an oracle that delivers synthesised answers rather than a directory that distributes attention. Using URL-level clickstream data on 45,386 US desktop panelists over October 2024–July 2025, we provide the first within-user empirical characterisation of how AI search reshapes web traffic. Four facts: (i) ChatGPT Search adoption was rapid and shock-driven, with search-enabled conversations growing roughly 1.7× over the panel window; (ii) AI search is absorptive—only 4.2% of ChatGPT conversations produce any outbound click vs. 31.7% of Google queries, 70.3% of ChatGPT users never click out at all (vs. 6.4% for Google), and within-user the referral-rate gap is ∼30 percentage points; (iii) AI search referrals are systematically tilted toward informational destinations (knowledge, academic, technical) and away from navigational and ad-supported content, and within a user-week ChatGPT reaches only ∼2 unique domains vs. Google's ∼14; (iv) exploiting two ChatGPT Search rollouts in a stacked event-study design, AI-search adoption causes a ∼9% reduction in Google queries, with the displacement concentrating in informational destinations such as Wikipedia, NYT, and Stack Overflow.

Governing AI Content Tools: The Efficiency–Diversity Trade-off in Platform Knowledge Ecosystems
Kai Zhu, **Kai Gu**, and Lorenzo Lupo

Presented at: 13th Wiki Workshop (Online), March 2026

Knowledge platforms increasingly deploy AI tools to accelerate content production, yet the ecosystem-level consequences of these decisions remain poorly understood. We study Wikipedia's Content Translation tool—one of the earliest large-scale AI content deployments on a major knowledge platform—using a matched sample of over half a million article pairs across ten language editions. We document an efficiency–diversity trade-off: machine translation (MT)–assisted articles increase information coverage for target-language readers by 50%, but reduce unique content contributions to the cross-language knowledge ecosystem by 48%, leaving total information supply essentially unchanged because MT substitutes translated source material for locally originated content rather than expanding the information frontier. The missing content is disproportionately culturally anchored—local context, regional perspectives, and community-specific details that human editors contribute but MT cannot generate. A difference-in-differences design exploiting a 2019 MT quality shock establishes causality: improved MT technology widens the content diversity shortfall rather than closing it. A calibrated simulation maps how aggregate coverage and content diversity co-evolve across MT adoption rates, providing a governance instrument for platforms navigating this trade-off.

Conference Presentations

ISMS Marketing Science Conference, Lisbon, Portugal	June 2026
Bocconi Marketing Research Camp	May 2026
13th Wiki Workshop (Online)	March 2026

Skills

Languages: English, Mandarin
Programming: R, Python, LaTeX, Markdown